Type MS · Explain how smoking damages health, including in pregnancy

Answer 1 of 6 · Question 2 part d · 4 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

Number				
2	(a) (i)	an answer that makes reference to three of the following: • warms air; • moistens air; • layer of (sticky) mucus; • removes/traps, pathogens/bacteria/microorganisms/dust / dirt (from entering lungs); • reference to 1st line of defence;	Mp3 must be linked to mp4 for mark	3
	(ii)	• prevents collapse; • to allow air to flow freely (to lungs/alveoli); • allows neck to bend;		2
	(iii)	B; (bronchi)		1
	(b)	D; (medulla oblongata)		1
	(c)	$17 \times 0.50 = 8.5\ dm^3$; 4×8.5; $= 0.34\ dm^3$;		3
	(d)	• reduced oxygen uptake by haemoglobin/forms carboxyhaemoglobin; • less oxygen to developing fetus; • lower rate of respiration/less energy released (by fetus); • less growth/lower birth mass;		4

Total 14 marks

Type MS · Explain how smoking damages health, including in pregnancy

Answer 1 of 6 · Question 2 part d · 4 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how smoking damages health, including in pregnancy

Answer 2 of 6 · Question 3 part c · 5 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

3(a)(i)	B 3.2 - 3.5 kg;		**(1)**

Question Number	Answer	Additional guidance	Mark
3(a)(ii)	(94 / 210) x 100 ; 45%; (2)	Ecf allow correct answer to incorrect figures used in calculation Allow 44.76%/ 44.8% for correct final answer	**(2)**

Question Number	Answer	Additional guidance	Mark
3(b)	smoking appears to lower birth mass; more higher birth mass babies born to non-smokers;	ORA for both mp's	**(2)**

Question Number	Answer	Additional guidance	Mark
3(c)	• cigarette smoke contains CO; • combines with haemoglobin instead of oxygen/forms carboxyhaemoglobin; • less oxygen to fetus; • less respiration in fetus; • lower growth rate/cell division; • causes lower birth mass;		**(5 max)**

Total 10 Marks

Type MS · Explain how smoking damages health, including in pregnancy

Answer 2 of 6 · Question 3 part c · 5 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

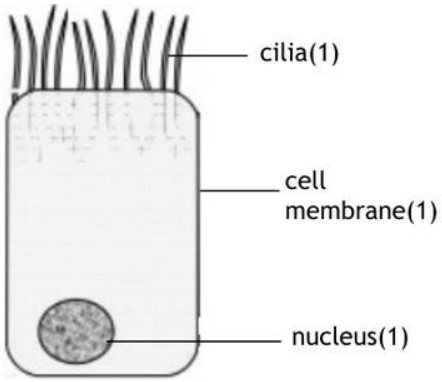
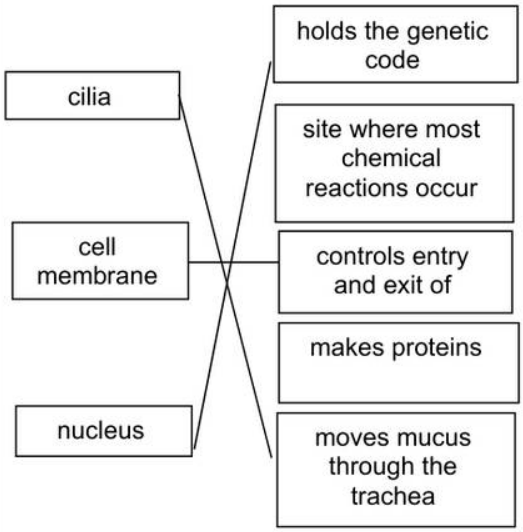
Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Explain how smoking damages health, including in pregnancy

Answer 3 of 6 · Question 2 part b.i · 1 mark

2020 NOV · Paper 01 (regional R) · find part b.i in this mark scheme extract

2 (a) (i)			3
(ii)		Reject more than one line from each structure	3
(b) (i)	Tar (1)		1
(ii)	• (cilia)burned/destroyed/reduced in number(1) • paralysed/cannot beat to and fro/move mucus(1)		2

Total for Question 2 = 9 Marks

Type MS · Explain how smoking damages health, including in pregnancy

Answer 4 of 6 · Question 3 part a · 2 marks

2024 JUN · Paper 02 · find part a in this mark scheme extract

Question number				
3	(a)	Smoking cigarettes; Causes (lung) cancer/emphysema/bronchitis/heart disease/eq ; OR lack of exercise; obesity/heart disease/high blood pressure/ type ii diabetes/eq;	Allow drugs or named drug and their effect on health for 2 marks.	(2)
	(b)	Any two from: • production of bile reduced/affected; • less fat emulsified/fat not emulsified; • lipase less active/lipase unable to digest as much (fat)/slower lipase activity;	Allow bile not produced Allow description of emulsification	(2)
	(c)	blurred/misty vision/blindness/less light passing through the lens / less light reaching the retina;		(1)
	(d)	cerebrum/cerebellum/ cerebral cortex/cerebral hemispheres/hypothalamus/medulla (oblongata);	Allow any part of the brain that is affected by alcohol	(1)
	(e)	Any three from: • reduces release of neurotransmitters; • reduced diffusion across synapse; • less neurotransmitters bind to receptors on post-synaptic membrane; • slows/ fewer transmission of impulses; • slower reaction /reflexes;		(3)
	(f)	Any two from • increased heart rate; • increased breathing rate; • pupil dilation; • increase in blood sugar/glucose levels/ increased breakdown of glycogen to glucose; • increase in blood flow to brain/muscles; • increase in (aerobic) respiration (in muscles);		(2)

Total for Question 3 = 11 marks

Type MS · Explain how smoking damages health, including in pregnancy

Answer 5 of 6 · Question 4 part a.iii · 2 marks

2025 JUN · Paper 02 (regional R) · find part a.iii in this mark scheme extract

4	(a)	(i)	• decreases; • from 82% to 72%/by 10%		2
		(ii)	79:21; 3.76:1;	accept 3.7612:1 accept 3.762:1 accept 3.8/4:1	2
		(iii)	any two from • passive smoking/ex- smokers already have damaged lungs; • (air) pollution; • genetics/heredity;		2
		(iv)	tar;		1
	(b)	(i)	bronchiole;		1
		(ii)	alveolus;		1
		(iii)	any four from • damaged alveoli/alveoli walls broken; • less surface area (for gas exchange); • difficulty breathing/shortness of breath; • less oxygen passing into the blood/ less carbon dioxide removed from blood/less gaseous exchange; • reduced (aerobic) respiration; • fatigue/less energy/not very active;		4
				Total for Question 4 = 13 marks	

Type MS · Explain how smoking damages health, including in pregnancy

Answer 6 of 6 · Question 4 part a.iv · 1 mark

2025 JUN · Paper 02 (regional R) · find part a.iv in this mark scheme extract

4	(a)	(i)	• decreases; • from 82% to 72%/by 10%		2
		(ii)	79:21; 3.76:1;	accept 3.7612:1 accept 3.762:1 accept 3.8/4:1	2
		(iii)	any two from • passive smoking/ex- smokers already have damaged lungs; • (air) pollution; • genetics/heredity;		2
		(iv)	tar;		1
	(b)	(i)	bronchiole;		1
		(ii)	alveolus;		1
		(iii)	any four from • damaged alveoli/alveoli walls broken; • less surface area (for gas exchange); • difficulty breathing/shortness of breath; • less oxygen passing into the blood/ less carbon dioxide removed from blood/less gaseous exchange; • reduced (aerobic) respiration; • fatigue/less energy/not very active;		4
				Total for Question 4 = 13 marks	